



Encryption in Google Messages

Google Messages is testing end-to-end encryption for RCS-enabled group chats. <https://digit.in/nov22-03>



OxygenOS 13 roadmap released

OnePlus has released the roadmap for the OxygenOS 13 update. <https://digit.in/nov22-04>

delivery via manual expression. This means that participants were required to pump out breastmilk using their hands by holding them in a C-cup shape to apply pressure and expel milk. Breastpumps were not used predominantly because they are made of plastic, which could have impacted the study adversely.

As a part of the study, the MPs were classified on the basis of size, colour and shape. The findings showed that most of these MPs were in the form of irregular fragments and spheres, a majority spanned 4-10-mm, and were coloured blue or yellow. MPs in the form of film or fibres were not found to be present in breastmilk.

This is alarmingly because of obvious reasons. An infant's primary source of nutrition is breastmilk, and when it is contaminated with MPs, it can cause not just nutritional deficits, but also physiological damage. Although there were no specific links between the participants, it is believed that mothers are exposed to plenty of MPs in daily life. The human body can ingest MPs by simply breathing, using personal care products that are available in plastic packaging, ingesting food and beverages, and even through tactile exposure in some cases. It is estimated that 39,000 - 52,000 MPs are ingested per person per annum.

WHY IS THIS CONCERNING?

Breastmilk has been found to contain a variety of contaminants like polychlorinated bisphenyls (PCBs), polybrominated diphenyl esters (PBDEs), organochlorine pesticides, per- and polyfluoroalkyl substances (PFASs), phthalates and phthalate metabolites, phenols, and even metals. While the impact of MPs on infant health and development is still being studied, it has been proven that substances such as PCB can lead to cognitive impairment on exposure in early life. It can also impact the endocrine system.

Similarly, PBDEs are known neurotoxins and can adversely affect

behavioural development, motor skills and cognitive development.

These contaminants are known to interact with MPs in the breastmilk, which enhancing their sorption capabilities. This means that the risk posed to infants increases not only because of the MPs themselves,



but also because of the additional toxins they may have absorbed from the breastmilk. With infants being especially vulnerable to harmful substances, this poses a large threat to their health.

WHAT ABOUT THE ADULTS?

This isn't the first time that scientists have reported finding MPs within the human body. In March 2022, a study conducted by the Dutch National Organisation for Health Research and Development and Common Seas found MPs in human blood for the first time, with the particles present in 80% of participants tested. MPs have also previously been detected in human stool (including the stool of infants).

Elaborating on the need for such studies, Professor Dick Vethaak, an ecotoxicologist at Vrije Universiteit Amsterdam, said, "The big question is what is happening in our body? Are the particles retained in the

body? Are they transported to certain organs, such as getting past the blood-brain barrier?" And are these levels sufficiently high to trigger disease? We urgently need to fund further research so we can find out."

Similarly, a study conducted in 2020 revealed the presence of MPs in the placentas of unborn babies. The findings showed that out of the 6 placentas studied, 4 contained MPs. The researchers explained "Due to the crucial role of placenta in supporting the foetus's development and in acting as an interface with the external environment, the presence of potentially harmful plastic particles is a matter of great concern. Further studies need to be performed to assess if the presence of microplastics may trigger immune responses or may lead to the release of toxic contaminants, resulting in harm."

While studies on how MPs impact foetal growth are underway, experiments on rats have shown that MPs in the placenta of pregnant rats can enter the lungs, brain, muscle tissue, kidney and other organs in the baby rat.

WHAT CAN BE DONE?

MPs have been discovered at the summit of Mount Everest, as well as deep oceans, demonstrating that



plastic pollution is inescapable. We inhale and ingest these particles daily, and with plastic production set to double by 2040, the situation is likely to become more dire. While we can all practise living a greener lifestyle on an individual basis, the truth is that this is an issue that requires deep-rooted systemic changes for any positive impact to take place. **d**

